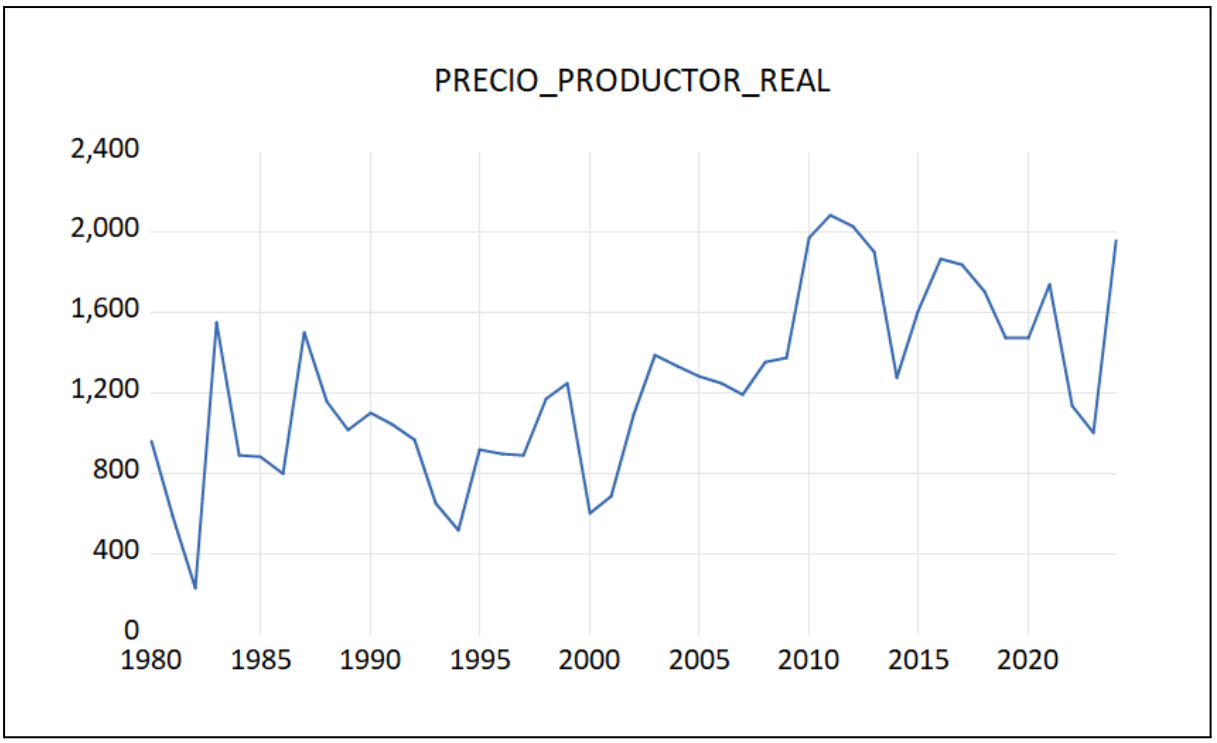

PRECIO_PRODUCTOR_REAL
2,400
2,000
1,600
1,200
800
400
0
1980
1985
1990
1995
2000
2005
2010
2015
2020

Null Hypothesis: PRECIO PRODUCTOR REAL has a unit root
Exogenous: Constant, Linear Trend
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-4.466497	0.0047
Test critical values:	1% level	-4.180911	
	5% level	-3.515523	
	10% level	-3.188259	

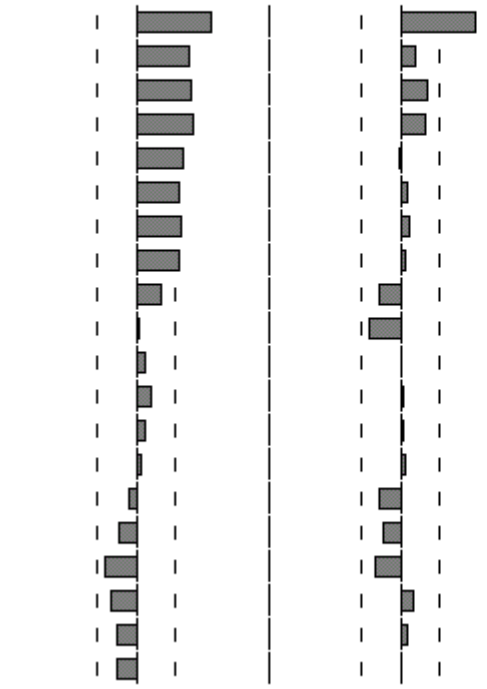
*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO PRODUCTOR REAL)
Method: Least Squares
Date: 11/05/26 Time: 12:25
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO PRODUCTOR REAL(-1)	-0.656128	0.146900	-4.466497	0.0001
C	477.4899	148.3321	3.219060	0.0025
@TREND("1980")	15.36929	4.988005	3.081249	0.0037

R-squared	0.328809	Mean dependent var	22.69169
Adjusted R-squared	0.296068	S.D. dependent var	390.7237
S.E. of regression	327.8197	Akaike info criterion	14.48855
Sum squared resid	4406097.	Schwarz criterion	14.61020
Log likelihood	-315.7481	Hannan-Quinn criter.	14.53366
F-statistic	10.04272	Durbin-Watson stat	1.850384
Prob(F-statistic)	0.000282		

Date: 11/05/26 Time: 12:25
Sample: 1980 2024
Included observations: 45

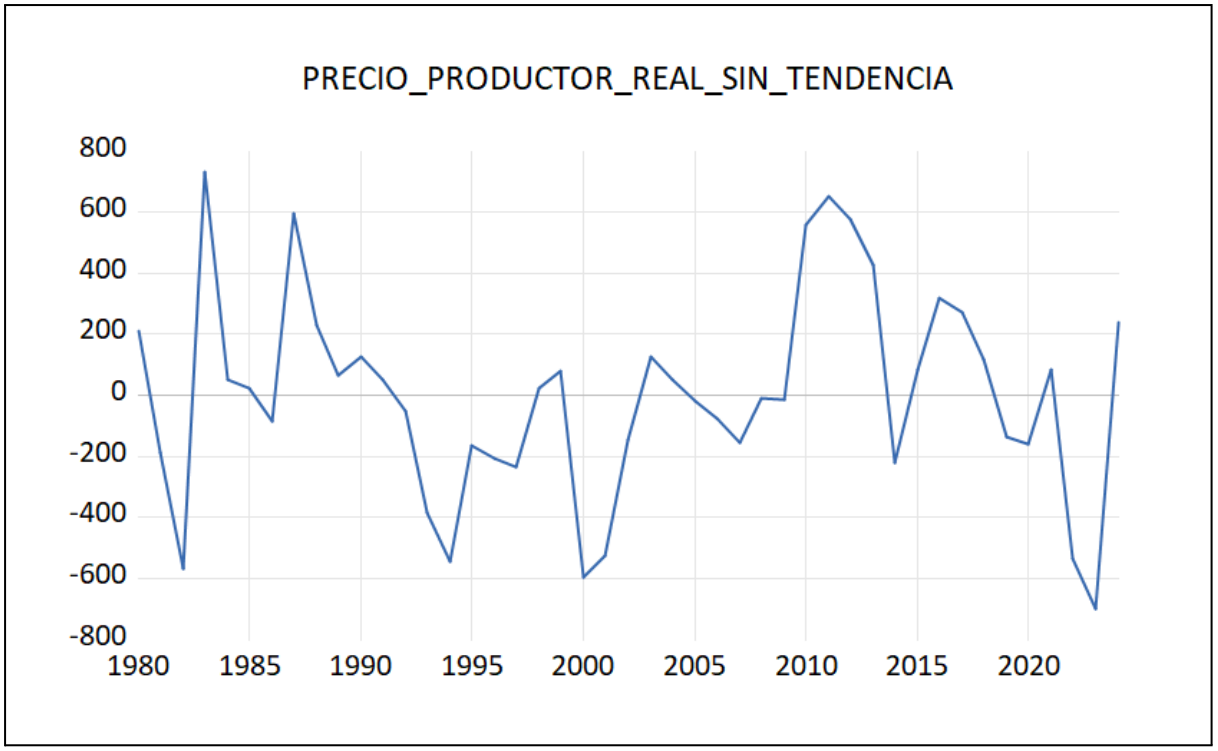


Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.588	0.588	16.606	0.000
		2	0.422	0.117	25.374	0.000
		3	0.429	0.218	34.641	0.000
		4	0.458	0.190	45.468	0.000
		5	0.368	-0.007	52.644	0.000
		6	0.335	0.060	58.738	0.000
		7	0.349	0.073	65.530	0.000
		8	0.337	0.033	72.019	0.000
		9	0.190	-0.166	74.139	0.000
		10	0.031	-0.248	74.197	0.000
		11	0.076	0.012	74.552	0.000
		12	0.114	0.030	75.391	0.000
		13	0.071	0.021	75.729	0.000
		14	0.039	0.041	75.831	0.000
		15	-0.054	-0.168	76.035	0.000
		16	-0.131	-0.132	77.284	0.000
		17	-0.244	-0.190	81.770	0.000
		18	-0.189	0.093	84.565	0.000
		19	-0.148	0.051	86.349	0.000
		20	-0.141	0.012	88.025	0.000

Dependent Variable: PRECIO PRODUCTOR REAL
Method: Least Squares
Date: 11/05/26 Time: 12:26
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	752.6148	100.4303	7.493902	0.0000
@TREND	22.00324	3.931145	5.597159	0.0000

R-squared	0.421485	Mean dependent var	1236.686
Adjusted R-squared	0.408031	S.D. dependent var	445.1336
S.E. of regression	342.4837	Akaike info criterion	14.55375
Sum squared resid	5043689.	Schwarz criterion	14.63405
Log likelihood	-325.4594	Hannan-Quinn criter.	14.58369
F-statistic	31.32819	Durbin-Watson stat	1.301550
Prob(F-statistic)	0.000001		



Date: 11/05/26 Time: 12:27
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.339	0.339	5.5253	0.019
		2	0.050	-0.073	5.6493	0.059
		3	0.111	0.134	6.2699	0.099
		4	0.142	0.071	7.3151	0.120
		5	0.009	-0.070	7.3192	0.198
		6	-0.023	-0.004	7.3473	0.290
		7	0.062	0.060	7.5627	0.373
		8	0.049	-0.002	7.6999	0.463
		9	-0.158	-0.186	9.1598	0.423
		10	-0.393	-0.337	18.476	0.047
		11	-0.314	-0.154	24.604	0.010
		12	-0.175	-0.059	26.565	0.009
		13	-0.152	-0.015	28.086	0.009
		14	-0.068	0.102	28.398	0.013
		15	-0.078	-0.065	28.821	0.017
		16	-0.153	-0.128	30.526	0.015
		17	-0.286	-0.240	36.716	0.004
		18	-0.142	0.010	38.297	0.004
		19	-0.041	-0.049	38.435	0.005
		20	-0.001	-0.084	38.435	0.008

Dependent Variable: PRECIO PRODUCTOR REAL SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:28
Sample: 1980 2024
Included observations: 45
Convergence achieved after 11 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.338436	0.125622	2.694081	0.0100
SIGMASQ	98937.02	19980.65	4.951641	0.0000

R-squared	0.117280	Mean dependent var	5.94E-14
Adjusted R-squared	0.096751	S.D. dependent var	338.5695
S.E. of regression	321.7744	Akaike info criterion	14.43171
Sum squared resid	4452166.	Schwarz criterion	14.51200
Log likelihood	-322.7134	Hannan-Quinn criter.	14.46164
Durbin-Watson stat	1.870049		

Inverted AR Roots	.34

Date: 11/05/26 Time: 12:28
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.035	0.035	0.0593	
		2	-0.117	-0.119	0.7359	0.391
		3	0.053	0.063	0.8771	0.645
		4	0.137	0.120	1.8425	0.606
		5	-0.037	-0.035	1.9147	0.751
		6	-0.059	-0.032	2.1046	0.834
		7	0.067	0.051	2.3542	0.884
		8	0.101	0.077	2.9423	0.890
		9	-0.062	-0.046	3.1664	0.923
		10	-0.312	-0.300	9.0540	0.432
		11	-0.185	-0.229	11.191	0.343
		12	-0.038	-0.126	11.282	0.420
		13	-0.091	-0.096	11.826	0.460
		14	0.012	0.102	11.837	0.541
		15	-0.003	0.017	11.837	0.619
		16	-0.053	-0.049	12.042	0.676
		17	-0.247	-0.257	16.644	0.409
		18	-0.056	-0.063	16.891	0.462
		19	0.002	-0.044	16.892	0.531
		20	-0.004	-0.071	16.893	0.597

Dependent Variable: PRECIO PRODUCTOR REAL SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:29
Sample: 1980 2024
Included observations: 45
Convergence achieved after 18 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.186682	0.138473	1.348149	0.1848
MA(10)	-0.758067	0.253559	-2.989705	0.0047
SIGMASQ	65292.94	18778.81	3.476948	0.0012

R-squared	0.417454	Mean dependent var	5.94E-14
Adjusted R-squared	0.389713	S.D. dependent var	338.5695
S.E. of regression	264.4933	Akaike info criterion	14.23734
Sum squared resid	2938182.	Schwarz criterion	14.35779
Log likelihood	-317.3402	Hannan-Quinn criter.	14.28224
Durbin-Watson stat	1.974574		

Inverted AR Roots	.19			
Inverted MA Roots	.97	.79+.57i	.79-.57i	.30-.93i
	.30+.93i	-.30-.93i	-.30+.93i	-.79-.57i
	-.79+.57i	-.97		

Dependent Variable: PRECIO PRODUCTOR REAL SIN TENDENCIA
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 12:31
Sample: 1980 2024
Included observations: 45
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

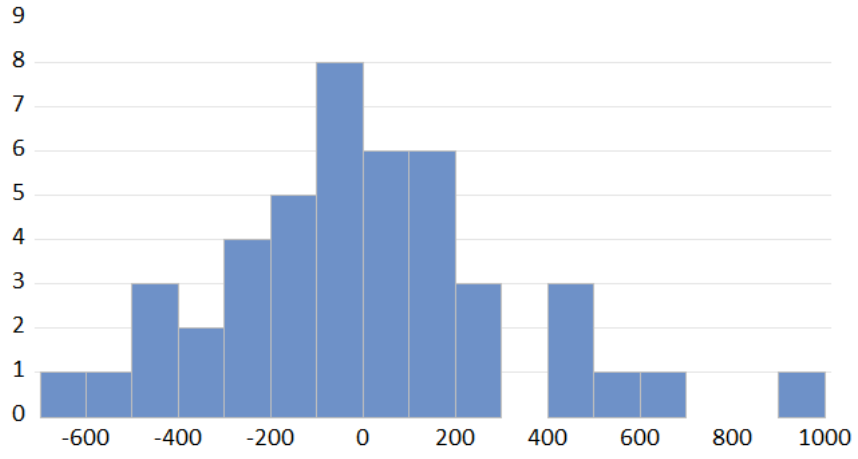
Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.377953	0.140547	2.689157	0.0102
SIGMASQ	97515.50	19075.29	5.112138	0.0000

R-squared	0.129963	Mean dependent var	5.94E-14
Adjusted R-squared	0.109729	S.D. dependent var	338.5695
S.E. of regression	319.4544	Akaike info criterion	14.41796
Sum squared resid	4388197.	Schwarz criterion	14.49825
Log likelihood	-322.4041	Hannan-Quinn criter.	14.44789
Durbin-Watson stat	1.923011		

Inverted MA Roots	-.38

Date: 11/05/26 Time: 12:31
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.014	0.014	0.0095	
		2	0.013	0.012	0.0172	0.896
		3	0.041	0.040	0.0999	0.951
		4	0.143	0.142	1.1499	0.765
		5	-0.033	-0.038	1.2068	0.877
		6	-0.034	-0.039	1.2704	0.938
		7	0.055	0.047	1.4410	0.963
		8	0.062	0.045	1.6594	0.976
		9	-0.069	-0.061	1.9369	0.983
		10	-0.302	-0.307	7.4378	0.592
		11	-0.204	-0.250	10.031	0.438
		12	-0.056	-0.079	10.232	0.510
		13	-0.117	-0.080	11.140	0.517
		14	0.008	0.114	11.145	0.599
		15	-0.043	0.008	11.275	0.664
		16	-0.056	-0.068	11.505	0.716
		17	-0.249	-0.257	16.207	0.439
		18	-0.056	-0.080	16.454	0.492
		19	-0.025	-0.021	16.504	0.557
		20	-0.002	-0.069	16.505	0.623

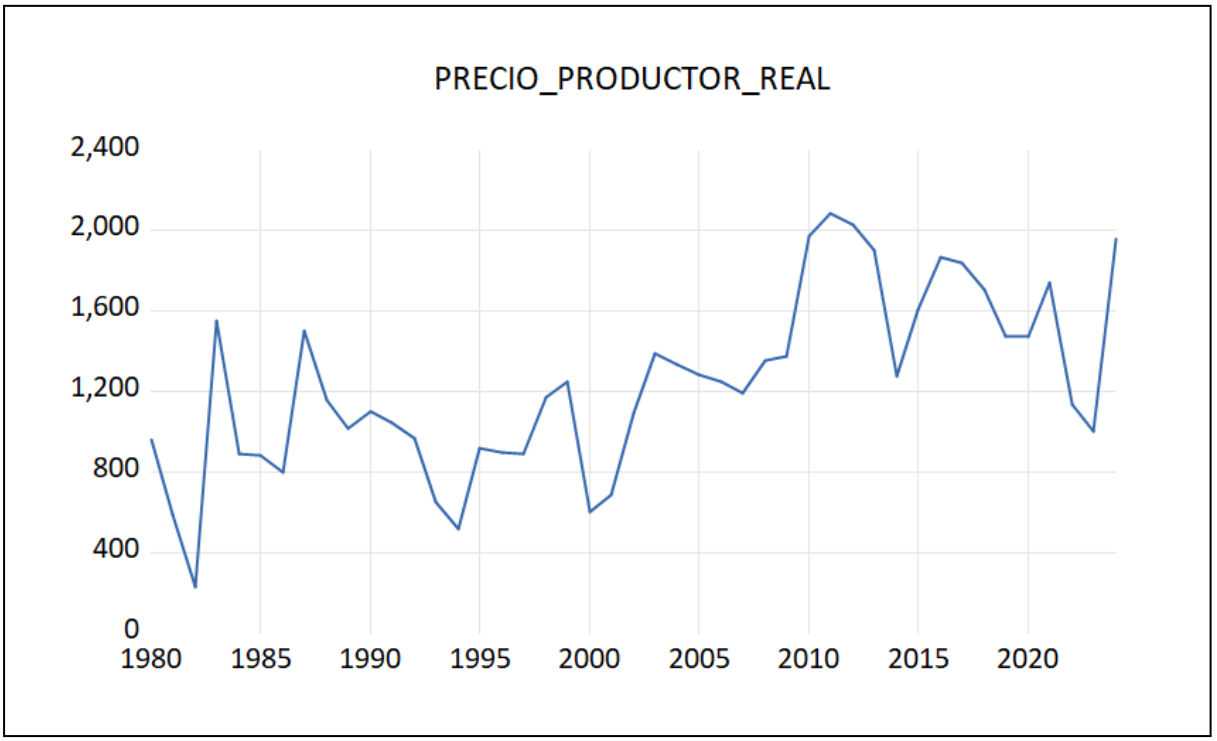


Series: Residuals
Sample 1980 2024
Observations 45

Mean	2.457488
Median	-10.63179
Maximum	911.0696
Minimum	-612.7970
Std. Dev.	315.7936
Skewness	0.431264
Kurtosis	3.527354
Jarque-Bera	1.916355
Probability	0.383591

Luego de realizar las estimaciones se ve como resulta significativo el rezago número diez en el modelo AR y MA, sin embargo incluirlo en la estimación genera problemas de significatividad en los coeficientes y de raíces cerca de la unidad, además si se diferencia la serie una vez, las estimaciones de los modelos arrojan una mayor varianza que los modelo en nivel, por lo tanto decido eliminar el componente MA(10) de ambas estimaciones y comparar los modelos AR(1) y MA(1) según el AIC, en donde gana el MA(1) por tener un menor criterio (14.41<14.43). Además se puede ver como el modelo MA(1) tiene raíces dentro del círculo unitario, y los residuos se comportan normalmente.

Incorporando una variable de intervencion:



Al igual que ocurrió con el precio internacional, el fuerte aumento del precio del cacao provocó que el gobierno Ghanes deba ajustar (aumentar) durante el 2024 el precio que establecido antes de la cosecha, además lo mismo ocurrió en el año 1983, con un fuerte aumento como incentivo a los productores de cacao como consecuencia de las reformas económicas (ERP).
Por lo tanto incorporo en la estimación ambas variables de intervención

Dependent Variable: PRECIO PRODUCTOR REAL
Method: Least Squares
Date: 11/05/26 Time: 15:35
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VI 1983	780.4922	340.4163	2.292758	0.0271
VI 2024	235.9633	343.4208	0.687097	0.4959
C	702.0902	100.8977	6.958439	0.0000
@TREND	23.27309	3.995232	5.825216	0.0000

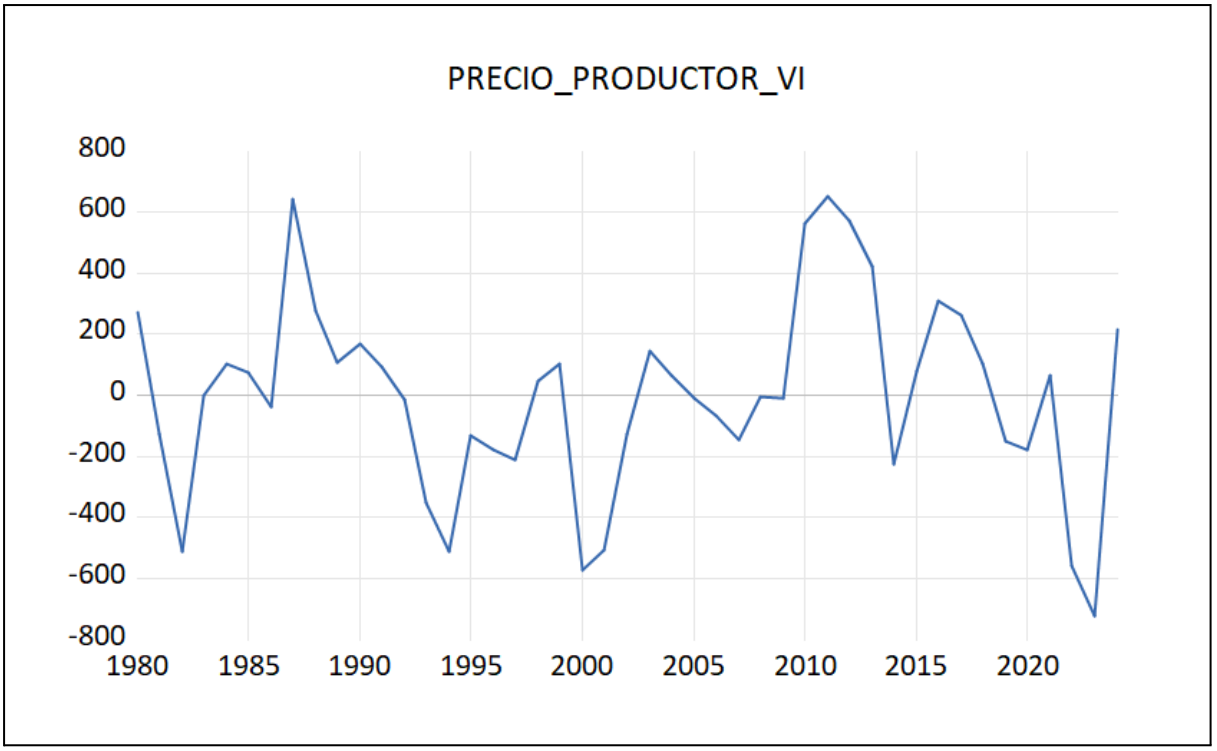
R-squared	0.493706	Mean dependent var	1236.686
Adjusted R-squared	0.456660	S.D. dependent var	445.1336
S.E. of regression	328.1151	Akaike info criterion	14.50929
Sum squared resid	4414040.	Schwarz criterion	14.66989
Log likelihood	-322.4591	Hannan-Quinn criter.	14.56916
F-statistic	13.32686	Durbin-Watson stat	0.974221
Prob(F-statistic)	0.000003		

Como sólo resulta significativa la variable de intervención de 1983, es la que dejo en el modelo.

Dependent Variable: PRECIO PRODUCTOR REAL
Method: Least Squares
Date: 11/05/26 Time: 15:36
Sample: 1980 2024
Included observations: 45

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VI 1983	788.8252	338.0554	2.333420	0.0245
C	691.6428	99.11655	6.978076	0.0000
@TREND	23.97790	3.836958	6.249195	0.0000

R-squared	0.487876	Mean dependent var	1236.686
Adjusted R-squared	0.463489	S.D. dependent var	445.1336
S.E. of regression	326.0465	Akaike info criterion	14.47630
Sum squared resid	4464867.	Schwarz criterion	14.59674
Log likelihood	-322.7167	Hannan-Quinn criter.	14.52120
F-statistic	20.00569	Durbin-Watson stat	1.048355
Prob(F-statistic)	0.000001		


PRECIO_PRODUCTOR_VI
800
600
400
200
0
-200
-400
-600
-800
1980
1985
1990
1995
2000
2005
2010
2015
2020

Null Hypothesis: PRECIO PRODUCTOR VI has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=9)

		t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic		-3.967477	0.0002
Test critical values:	1% level	-2.618579	
	5% level	-1.948495	
	10% level	-1.612135	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(PRECIO PRODUCTOR VI)
Method: Least Squares
Date: 11/05/26 Time: 15:38
Sample (adjusted): 1981 2024
Included observations: 44 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
PRECIO PRODUCTOR VI(-1)	-0.532803	0.134293	-3.967477	0.0003

R-squared	0.267960	Mean dependent var	-1.286211
Adjusted R-squared	0.267960	S.D. dependent var	329.9292
S.E. of regression	282.2853	Akaike info criterion	14.14618
Sum squared resid	3426454.	Schwarz criterion	14.18673
Log likelihood	-310.2159	Hannan-Quinn criter.	14.16122
Durbin-Watson stat	1.655973		

Date: 11/05/26 Time: 15:38
Sample: 1980 2024
Included observations: 45

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.462	0.462	10.275	0.001
		2	0.072	-0.180	10.531	0.005
		3	0.093	0.180	10.968	0.012
		4	0.046	-0.096	11.079	0.026
		5	-0.042	-0.019	11.174	0.048
		6	-0.051	-0.025	11.315	0.079
		7	0.035	0.085	11.383	0.123
		8	0.026	-0.042	11.423	0.179
		9	-0.191	-0.236	13.568	0.139
		10	-0.398	-0.282	23.163	0.010
		11	-0.286	-0.002	28.242	0.003
		12	-0.195	-0.112	30.672	0.002
		13	-0.157	0.028	32.308	0.002
		14	-0.048	0.009	32.465	0.003
		15	-0.094	-0.196	33.083	0.005
		16	-0.186	-0.143	35.613	0.003
		17	-0.214	-0.127	39.085	0.002
		18	-0.060	0.122	39.369	0.003
		19	-0.014	-0.170	39.385	0.004
		20	-0.017	-0.108	39.410	0.006

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:40
Sample: 1980 2024
Included observations: 45
Convergence achieved after 35 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.066811	0.236201	0.282856	0.7787
MA(1)	0.583150	0.225476	2.586313	0.0133
SIGMASQ	70102.35	13363.45	5.245828	0.0000

R-squared	0.293460	Mean dependent var	9.35E-14
Adjusted R-squared	0.259815	S.D. dependent var	318.5502
S.E. of regression	274.0614	Akaike info criterion	14.13996
Sum squared resid	3154606.	Schwarz criterion	14.26040
Log likelihood	-315.1490	Hannan-Quinn criter.	14.18486
Durbin-Watson stat	1.909410		

Inverted AR Roots	.07
Inverted MA Roots	-.58

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:40
Sample: 1980 2024
Included observations: 45
Convergence achieved after 7 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.464593	0.120781	3.846577	0.0004
SIGMASQ	77433.23	15884.10	4.874890	0.0000

R-squared	0.219575	Mean dependent var	9.35E-14
Adjusted R-squared	0.201425	S.D. dependent var	318.5502
S.E. of regression	284.6661	Akaike info criterion	14.18934
Sum squared resid	3484495.	Schwarz criterion	14.26964
Log likelihood	-317.2602	Hannan-Quinn criter.	14.21927
Durbin-Watson stat	1.695829		

Inverted AR Roots	.46

Date: 11/05/26 Time: 15:41
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.100	0.100	0.4854	
		2	-0.228	-0.240	3.0344	0.082
		3	0.055	0.115	3.1873	0.203
		4	0.045	-0.036	3.2921	0.349
		5	-0.067	-0.031	3.5287	0.474
		6	-0.081	-0.075	3.8809	0.567
		7	0.063	0.063	4.0998	0.663
		8	0.137	0.101	5.1754	0.639
		9	-0.068	-0.069	5.4471	0.709
		10	-0.333	-0.299	12.132	0.206
		11	-0.104	-0.089	12.809	0.235
		12	-0.035	-0.165	12.886	0.301
		13	-0.090	-0.053	13.416	0.340
		14	0.089	0.097	13.957	0.377
		15	0.009	-0.096	13.963	0.452
		16	-0.106	-0.125	14.778	0.468
		17	-0.184	-0.239	17.336	0.364
		18	0.038	0.088	17.447	0.424
		19	0.022	-0.088	17.488	0.490
		20	-0.053	-0.097	17.725	0.541

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:42
Sample: 1980 2024
Included observations: 45
Convergence achieved after 12 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(1)	0.358847	0.115699	3.101552	0.0034
AR(10)	-0.419358	0.127957	-3.277327	0.0021
SIGMASQ	61265.33	16870.37	3.631535	0.0008

R-squared	0.382526	Mean dependent var	9.35E-14
Adjusted R-squared	0.353122	S.D. dependent var	318.5502
S.E. of regression	256.2058	Akaike info criterion	14.05054
Sum squared resid	2756940.	Schwarz criterion	14.17099
Log likelihood	-313.1372	Hannan-Quinn criter.	14.09544
Durbin-Watson stat	1.722941		

Inverted AR Roots	.92-.28i	.92+.28i	.58+.74i	.58-.74i
	.03+.91i	.03-.91i	-.51-.74i	-.51+.74i
	-.84-.28i	-.84+.28i		

Date: 11/05/26 Time: 15:43
Sample: 1980 2024
Q-statistic probabilities adjusted for 2 ARMA terms

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.105	0.105	0.5274	
		2	-0.224	-0.237	2.9923	
		3	0.079	0.142	3.3031	0.069
		4	0.077	-0.009	3.6097	0.164
		5	-0.129	-0.096	4.4857	0.214
		6	-0.167	-0.137	5.9918	0.200
		7	0.007	-0.011	5.9943	0.307
		8	0.130	0.095	6.9633	0.324
		9	-0.072	-0.078	7.2671	0.402
		10	-0.121	-0.061	8.1545	0.419
		11	0.043	-0.017	8.2687	0.507
		12	0.046	-0.010	8.4020	0.590
		13	-0.049	-0.006	8.5619	0.662
		14	0.070	0.110	8.8963	0.712
		15	0.010	-0.076	8.9028	0.780
		16	-0.168	-0.174	10.950	0.690
		17	-0.261	-0.269	16.092	0.376
		18	0.064	0.096	16.411	0.425
		19	0.021	-0.079	16.448	0.492
		20	-0.176	-0.107	19.077	0.387

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:44
Sample: 1980 2024
Included observations: 45
Convergence achieved after 29 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
MA(1)	0.622026	0.153787	4.044721	0.0002
SIGMASQ	70237.87	13432.52	5.228941	0.0000

R-squared	0.292094	Mean dependent var	9.35E-14
Adjusted R-squared	0.275631	S.D. dependent var	318.5502
S.E. of regression	271.1176	Akaike info criterion	14.09728
Sum squared resid	3160704.	Schwarz criterion	14.17758
Log likelihood	-315.1888	Hannan-Quinn criter.	14.12721
Durbin-Watson stat	1.865293		

Inverted MA Roots	-.62

Date: 11/05/26 Time: 15:44
Sample: 1980 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.032	0.032	0.0479	
		2	0.008	0.007	0.0514	0.821
		3	0.027	0.026	0.0873	0.957
		4	0.067	0.066	0.3212	0.956
		5	-0.069	-0.074	0.5742	0.966
		6	-0.040	-0.037	0.6607	0.985
		7	0.030	0.030	0.7101	0.994
		8	0.055	0.053	0.8803	0.997
		9	-0.063	-0.056	1.1128	0.997
		10	-0.307	-0.312	6.7987	0.658
		11	-0.148	-0.162	8.1688	0.612
		12	-0.042	-0.034	8.2794	0.688
		13	-0.150	-0.124	9.7574	0.637
		14	0.095	0.152	10.375	0.663
		15	-0.073	-0.120	10.750	0.706
		16	-0.067	-0.129	11.081	0.747
		17	-0.198	-0.224	14.040	0.596
		18	0.031	0.029	14.115	0.659
		19	-0.022	-0.002	14.153	0.719
		20	-0.031	-0.147	14.236	0.770

Dependent Variable: PRECIO PRODUCTOR VI
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 11/05/26 Time: 15:44
Sample: 1980 2024
Included observations: 45
Convergence achieved after 28 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(10)	-0.464290	0.163360	-2.842135	0.0069
MA(1)	0.540656	0.158322	3.414922	0.0014
SIGMASQ	56960.11	16349.65	3.483873	0.0012

R-squared	0.425917	Mean dependent var	9.35E-14
Adjusted R-squared	0.398580	S.D. dependent var	318.5502
S.E. of regression	247.0398	Akaike info criterion	13.98300
Sum squared resid	2563205.	Schwarz criterion	14.10344
Log likelihood	-311.6175	Hannan-Quinn criter.	14.02790
Durbin-Watson stat	1.916948		

Inverted AR Roots	.88-.29i	.88+.29i	.54-.75i	.54+.75i
	-.00+.93i	-.00-.93i	-.54+.75i	-.54-.75i
	-.88+.29i	-.88-.29i		
Inverted MA Roots	-.54			

Nuevamente, al igual que antes de incorporar la variable de intervención aparece el rezago 10 casi significativo, si se lo incorpora, por un lado mejora el AIC, pero las raíces quedan muy cerca de la unidad, también rompe con la parsimonia de un modelo que de por si ya posee pocas observaciones (45), por lo que incluirlo limitaría mucho los grados de libertad, además de que económicamente hablando implicaría suponer que el precio del productor está determinado por lo ocurrido hace 10 años, lo cual no le encuentro mucho sentido, por lo tanto decido no incluirlo en el modelo, a pesar de que los residuos no queden completamente ruido blanco; además diferenciar la serie no es una opción debido a que aumenta la varianza en las estimaciones y el ADF fue rechazado anteriormente.
Por lo tanto nuevamente comparo los modelos AR(1) y MA(1), ambos sin el componente del rezago 10, y el que tiene menor AIC es el modelo **MA(1),** que posee **residuos aproximadamente ruido blanco**, debido a la presencia de la autocorrelación en el décimo rezago, lo cual debe ser tenido en cuenta para la aplicación de próximos modelos como por ejemplo el Modelo de Función de Transferencia**.**

